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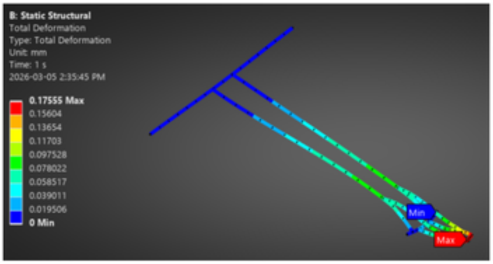
FSAE Modular Push Bar/Rear Jack



Task

- Create a mechanism which can jack the car up to full wheel rebound
- Allow 2 people to push the car without touching it, per rules
- Carry a fire extinguisher to be easily removed

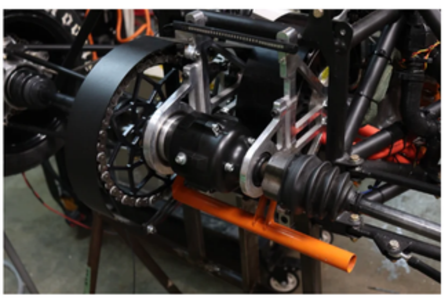
Process

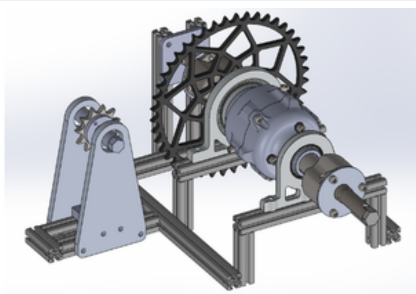
- Iteratively performed FEA using Ansys Workbench to modify handle length, handle thickness, and claw length
- Performed mesh refinement in Ansys SpaceClaim and Mechanical to validate calculations
- Reduced fire extinguisher mount weight by 83%, and drastically reduced time to access fire extinguisher

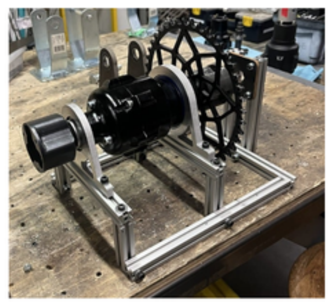
Result

- Improved balance, reduced time to access fire extinguisher, made the mechanism modular, and modified angle of push as per feedback from previous iterations
- Passed mechanical inspection at FSAE Hybrid+Electric, and was able to comfortably push/jack the car

Static Differential Test Bench







Task

- Create a test bench to hold the differential in place to be torqued
- Be able to lock one side of the diff to measure breakaway torque
- Aid in differential assembly

Process

- Designed the test bench in SolidWorks with a locking plate and a mock motor output shaft to apply torque through a chain
- Conduct tests to measure the breakaway torque based on fastener torque, oil fullness, and component wear
- Used DFMA to allow for easy installation/removal of the differential to aid in the assembly of the differential itself

Result

- Use breakaway torque values to calculate preload
- Standardize differential fastener torques for more consistent performance
- Applying preload and friction coefficient values to predict torque bias ratio based off of component geometry

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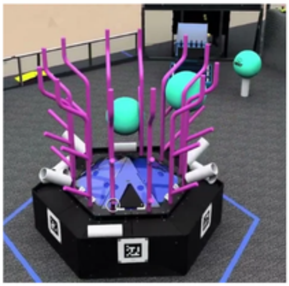
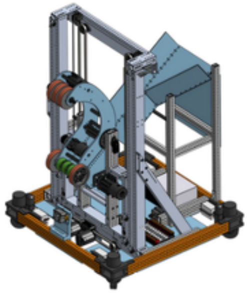
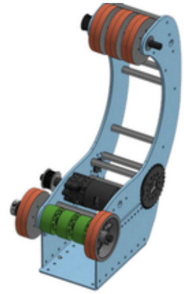
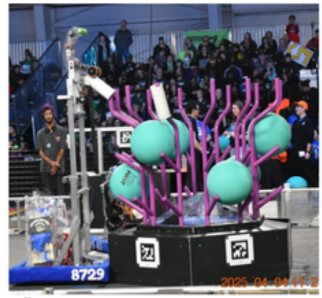
FRC Robot (Spark Youth Robotics 8729)









Task

- Create a rules-compliant robot to intake tubes and balls from a raised human-operated platform
- Output game items onto a rack (pictured above)
- Perform consistently across 40+ matches

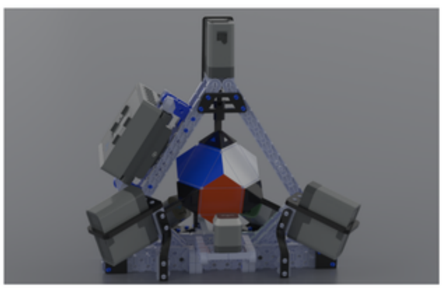
Process

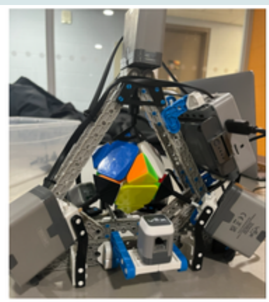
- Designed all robot architecture in Onshape, including drivetrain, intake, outtake, and elevator systems.
- Designed and assembled a custom claw with 2 Brushless Motors connected to 2 flywheel shafts for outtaking 2 types of game elements, polycarbonate side panels, and aluminum extrusion support beams.
- Led manufacturing & assembly including the cutting & drilling aluminum extrusions, 3D printing custom parts, assembling swerve drive modules, and assembling all systems together

Result

- Placed 18/50 at the provincial championship, with 148 teams competing in the province
- Performed consistently across 40+ matches, and improved robot performance by 52% across the season
- Led a team of 30+ students to complete design, prototyping, and manufacturing of a multi-mechanism robot within 14 weeks

Rubik's Skewb Solving Robot





Task

- Create a robot to receive information about the Skewb's state, and then autonomously solve it within 1 minute, with at least an 80% solve rate
- Use sensors to verify the cube's orientation and correct placement at all times

Process

- Designed 10+ custom 3D-printed parts in SolidWorks to stabilize motors and support sensors
- Tuned and repositioned VEX IQ Components to increase consistency by 40% to account for lighting and account for human error
- Wrote startup and emergency shutdown procedures, making use of a distance, optical, and touch sensor to ensure proper skewb installation

Result

- Increased motor accuracy by 64% with 3D-printed supports
- Solved the cube in as fast as 13 seconds
- Achieved an 80% solve rate
- Reached 90% sensor consistency in a variety of environments